

Introduction to Geothermal Project Feasibility

Sharing session exclusively prepared for PT Indonesia Power



Jacobs

Challenging today.
Reinventing tomorrow.

One of the largest Engineering and Consulting firms in the world

- 400+ offices in 40+ countries
- 55,000 permanent staff globally

1262+ Geothermal projects / assignments over last 40 years in 38 countries involving more than 4000MW:

- 391 Geothermal assignments in Indonesia at 41 locations
- 238 Geothermal assignments in ASEAN
- 633 Geothermal assignments in 33 other countries

World ENR Rank #1 Best Engineering and Design Firm 2018, 2019, 2020, 2021

Jacobs Presenters



Gandi Iswara
Project Manager / Sr. Process Engineer,
Jacobs Indonesia
16 years experience, (7 years geothermal)



Ridha Hendri, MT
Geologist, Jacobs Indonesia
10 years experience
Drilled 50+ geothermal wells



Eddiyanto
Geothermal Engineer
25+ years experience
Well Testing, Business
Planning, Construction Lead

Agenda

Session	Time	Presenter
Session-1 Introduction : Why Geothermal	1.30 – 1.40	Gandi
Session-2 Geoscience and Exploration	1.40 – 2.10 Q&A (15min)	Ridho
Break	5 min	
Session -3 Geothermal Project Development Steps	2.30 – 2.50 Q&A (10min)	Eddiyanto
Break	5 min	
Session -4 Indicative Cost	3.05 – 3.45 Q&A and wrap-up (15min)	Gandi
Session -5 Beyond Geothermal Electricity: Green Hydrogen		

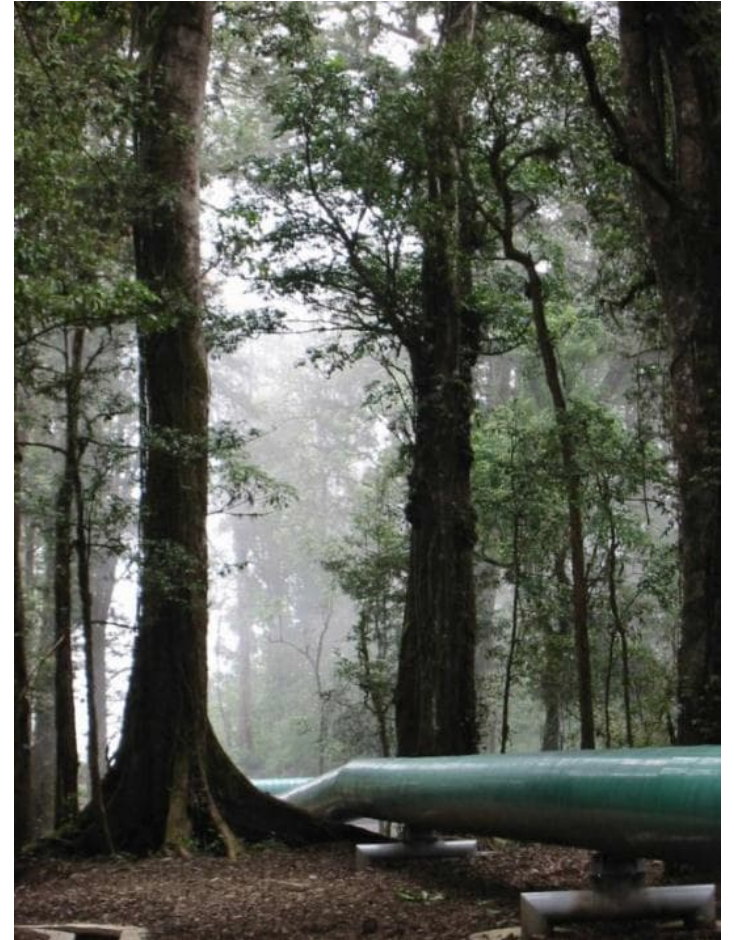
Why Geothermal

Green, base load, local

Geothermal is “Green”

- Sustainable – long term
- Low Carbon
 - Only small emissions
 - Attracts carbon credits
- Relatively Low Visual Impact
- Multiple Land Uses

Photo: Jacobs designed geothermal project
in Indonesia
*Green Energy
without disturbing the rainforest !*



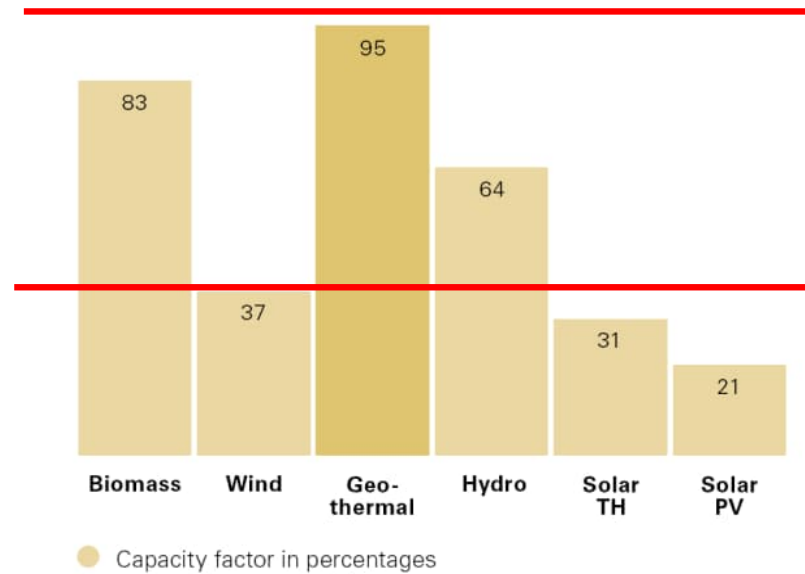
Baseload, high Capacity (% of time generating)

Better than other 'renewable', comparable to Thermal plant

Thermal Base
Load

Typical Wind &
Solar

- Dispatchable
- Firm capacity
- High reliability
- High availability



Source: U.S. Energy Information Agency

Solar has capital cost of about \$1000/kW, but capacity factor of about 25% at best.

Geothermal has capital cost of ~\$4000-5000/kW but capacity factor of near 100%

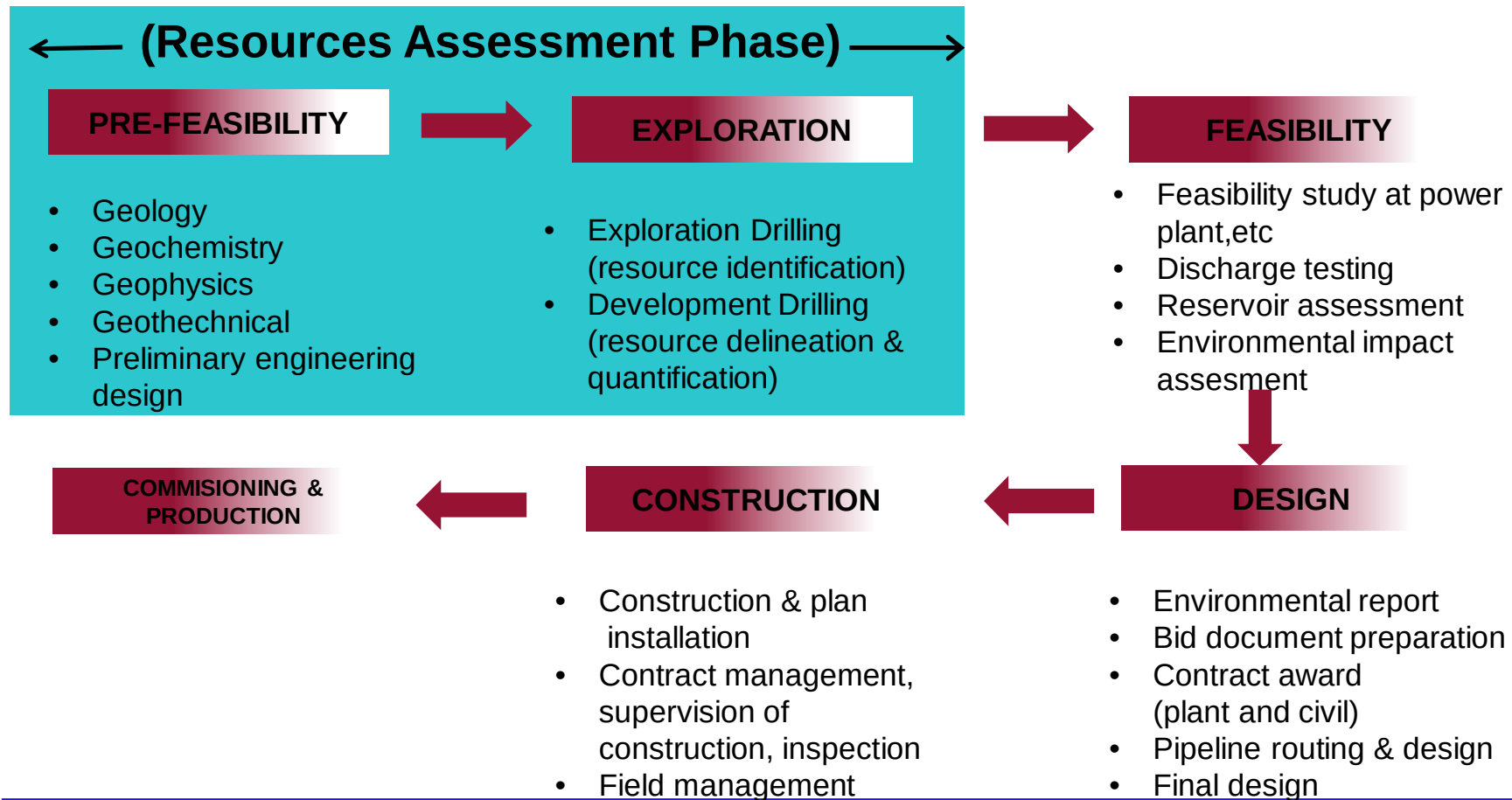
Security of fuel supply – once proven

- Owner controls the fuel supply
- Future cost largely known
- But requires ongoing management
 - Monitoring
 - Science interpretation
 - Drilling
 - Change to steamfield to bring fuel to plant

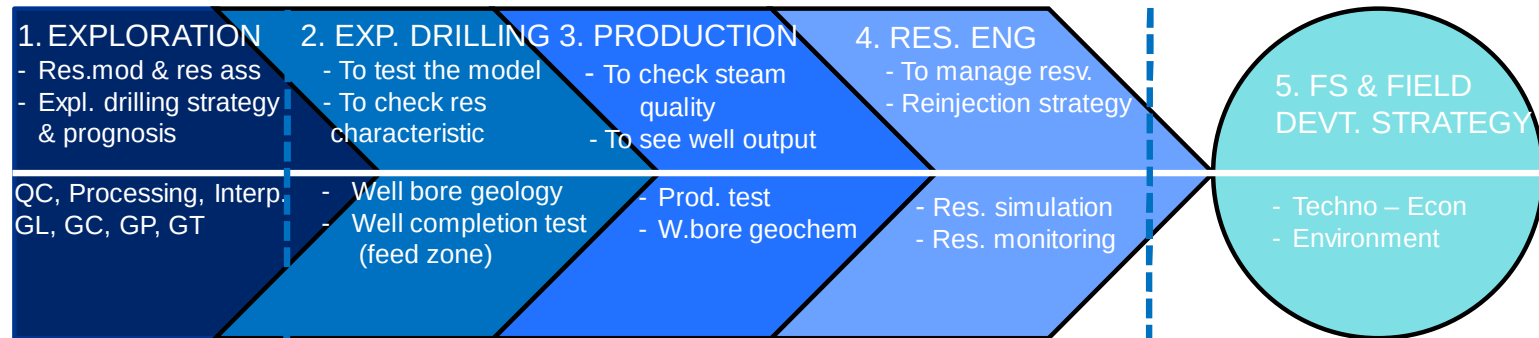


Geoscience & Exploration

Geothermal Development Stages



← EXPLORATION (Resources Ass & FS) →



← Project (Pre Production) →

PIC :

- Geologist
- Geochemist
- Geophysicst
- Civil Engineer

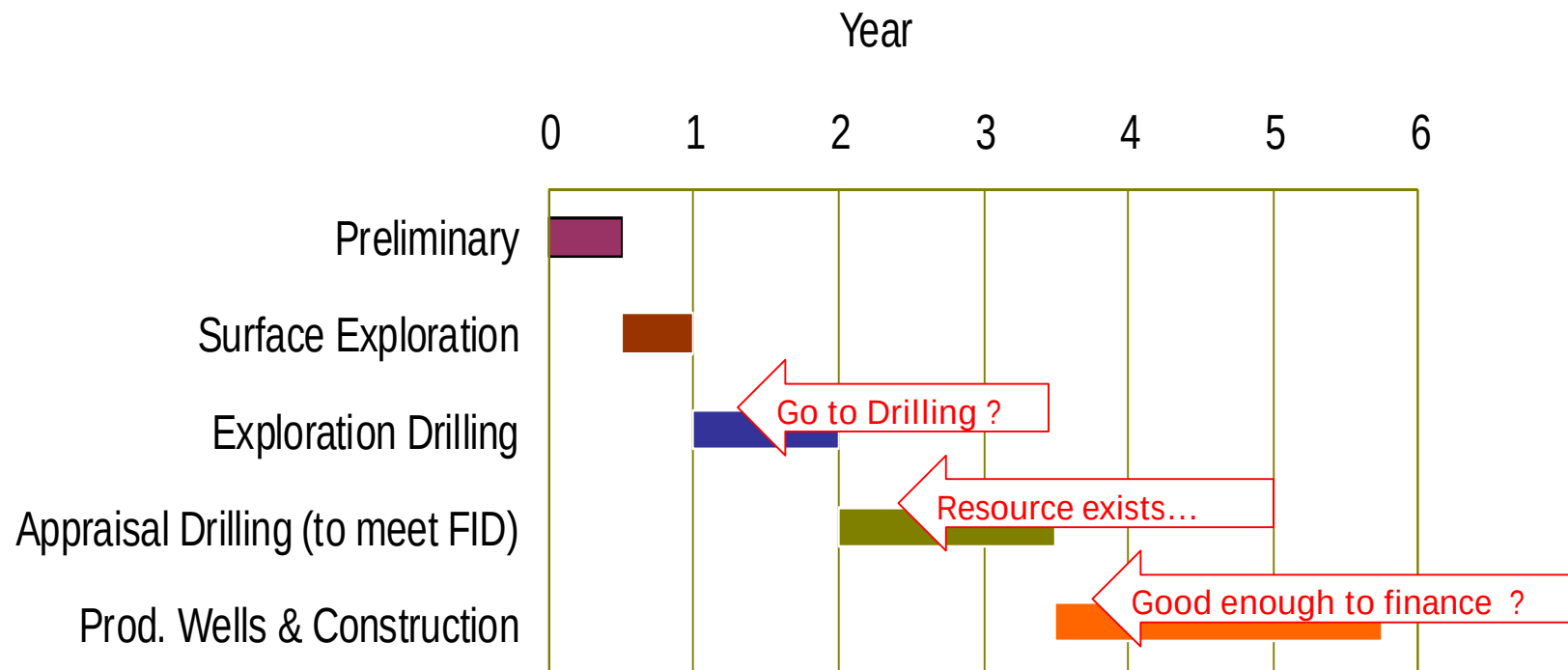
PIC :

- Geologist
- Geochemist
- Geophysicst
- Reservoir Engineer
- Drilling Engineer

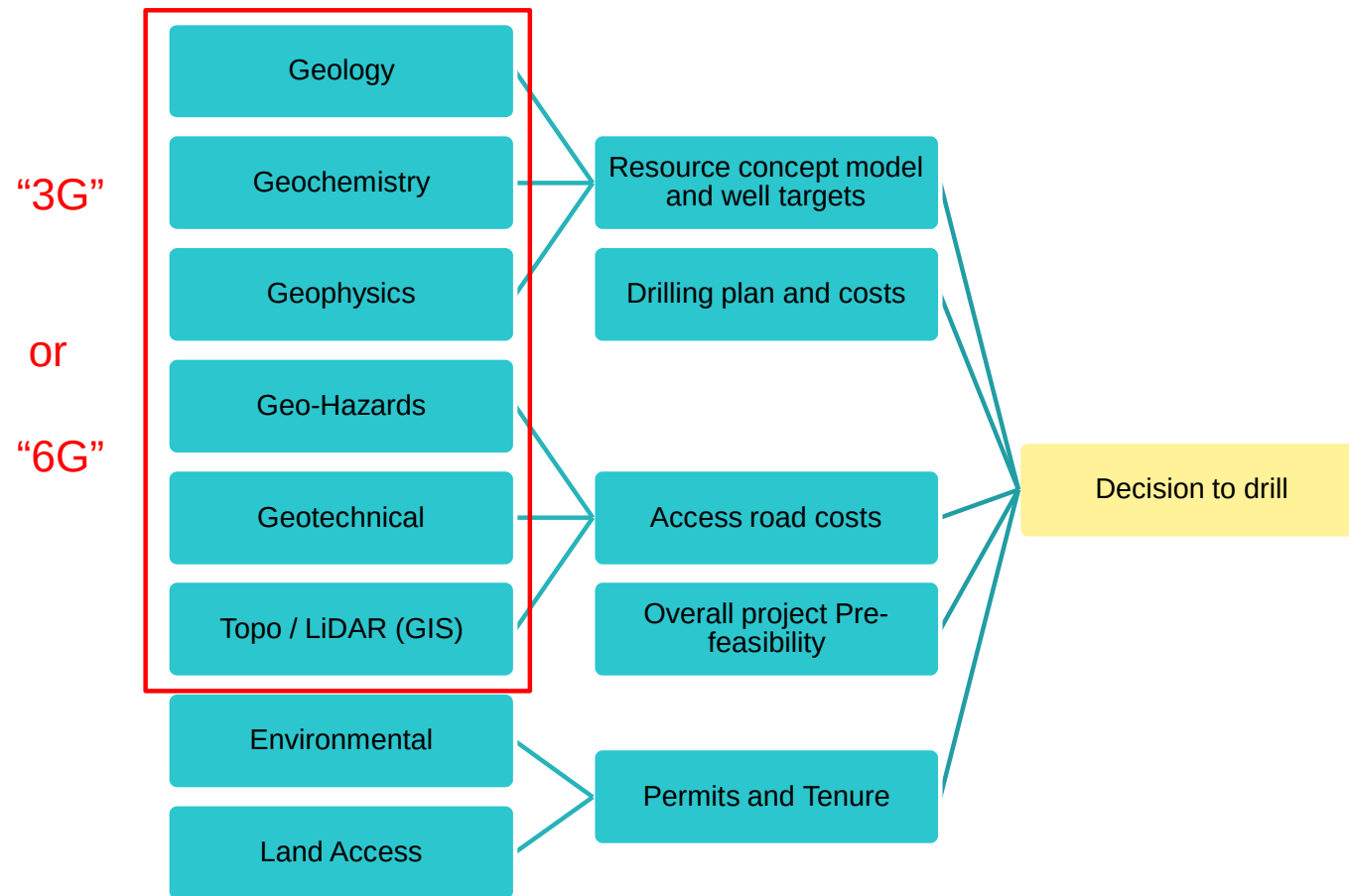
PIC :

- Geologist
- Geochemist
- Reservoir Engineer
- Environmental Engineer
- Civil, Electrical & Mechanical

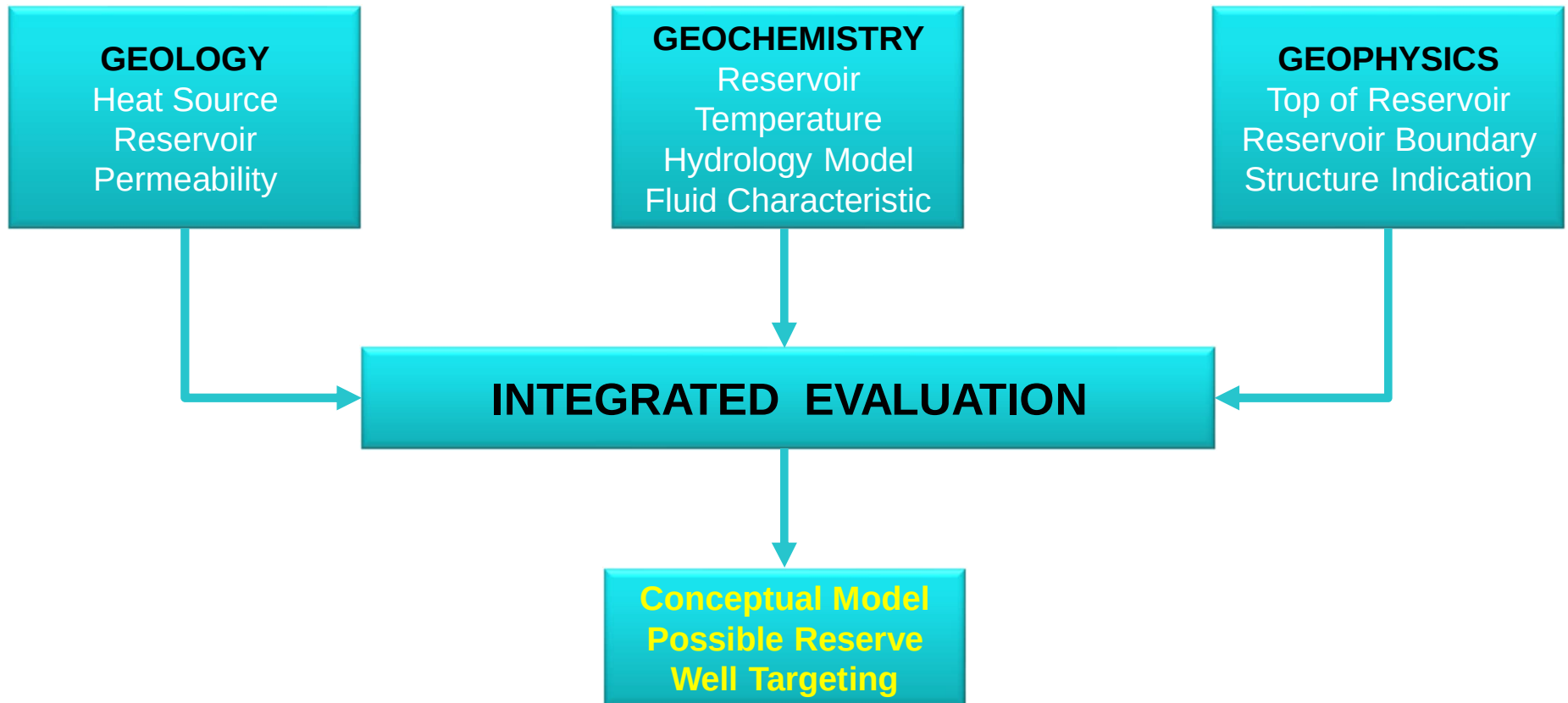
Key decision points, and opportunity for delay...



Proceeding with exploration drilling



3G Deliverable

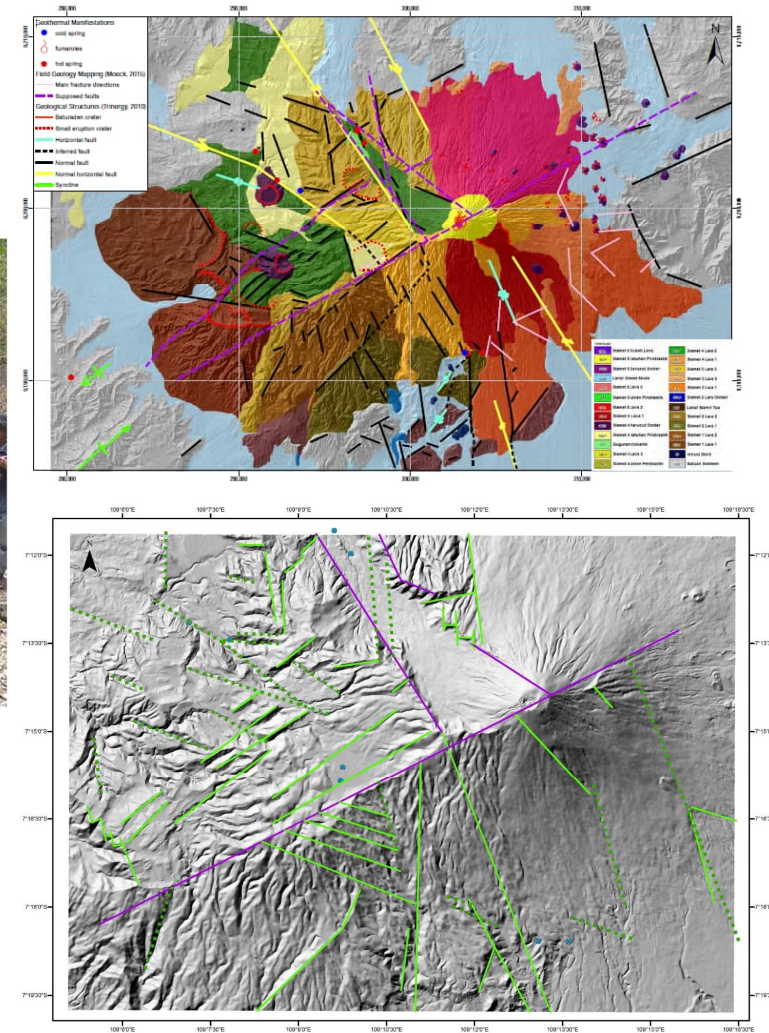


GEOLOGY :

- Remote Sensing
 - Tentative Geological Map
 - Lineament Structural Analysis
- Geological Mapping:
 - Rock Sampling
 - Structural Analysis
 - Thermal Manifestation and Alteration Mapping
- Laboratories:
 - Petrography
 - XRD
- Geological Model

Outcomes:

- Heat source
- Reservoir rocks
- Permeability



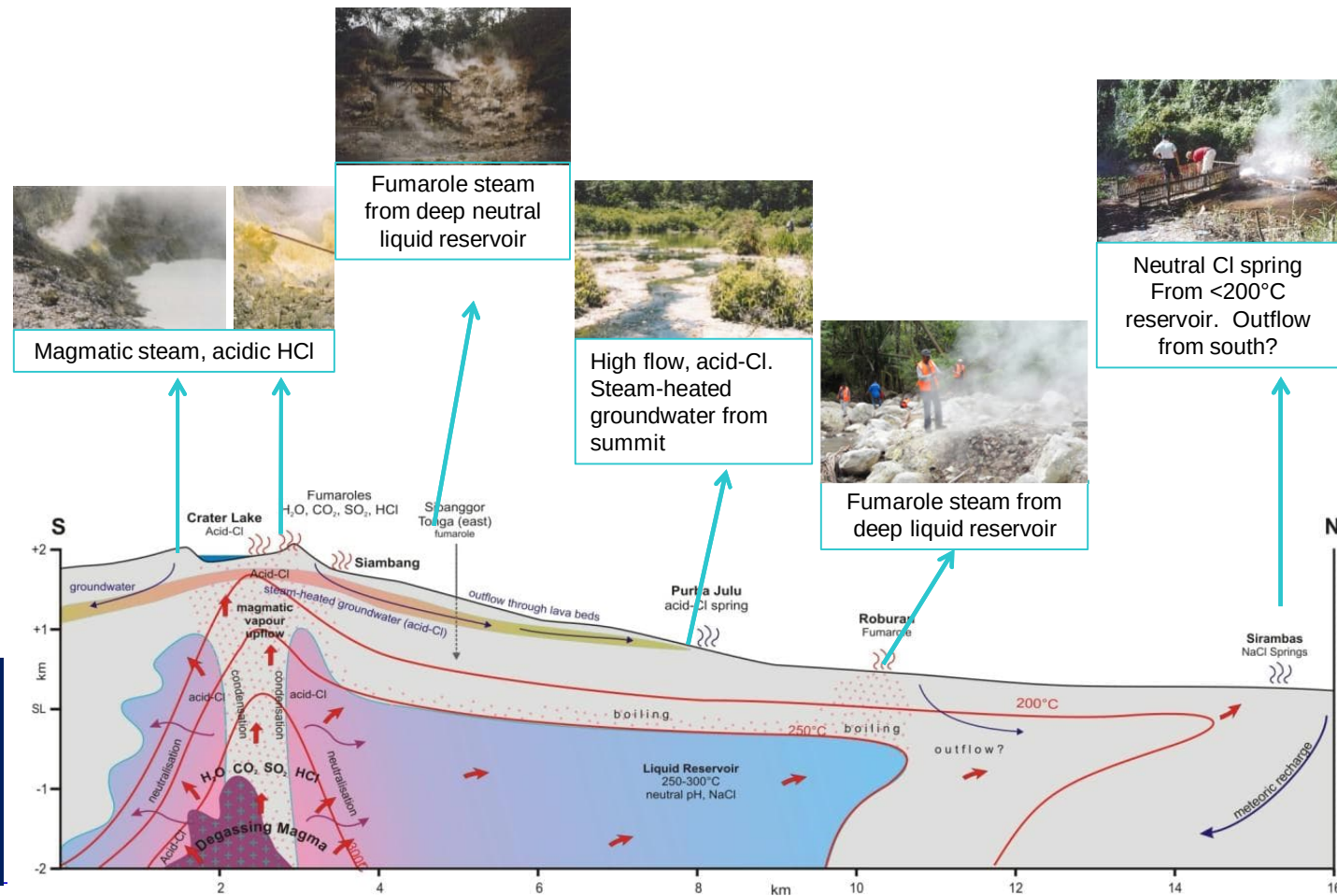
GEOCHEMISTRY :

- Water & Gas sampling:
 - Location
 - Temperature
 - Conductivity
 - pH
 - Flow rate
- Laboratory analysis:
 - Chemical Composition
 - Ion balance
 - Anion-cation
 - Isotope
- Interpretation

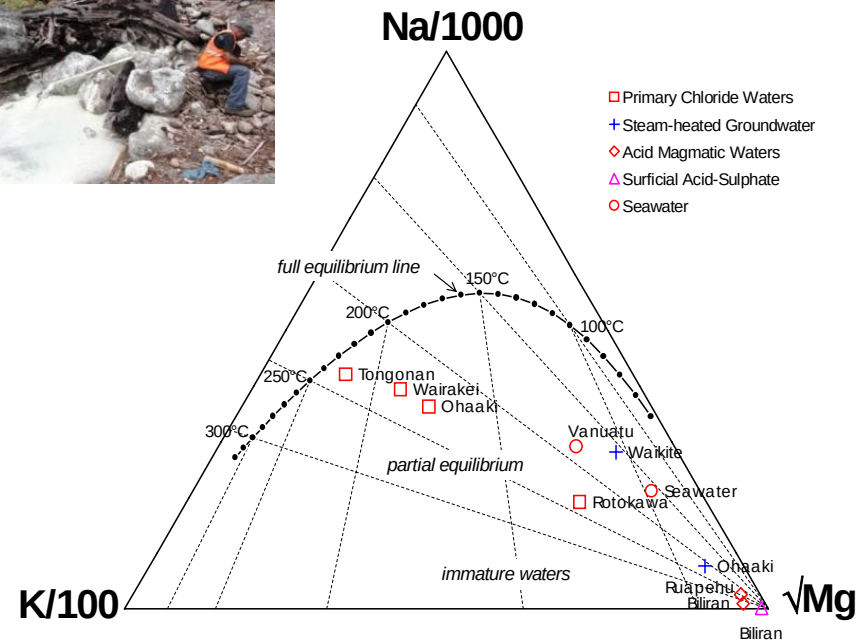
Outcomes:

- Reservoir temperature
- Hydrology model
- Fluid characteristic

Surface Thermal Activity – indicate system hydrology



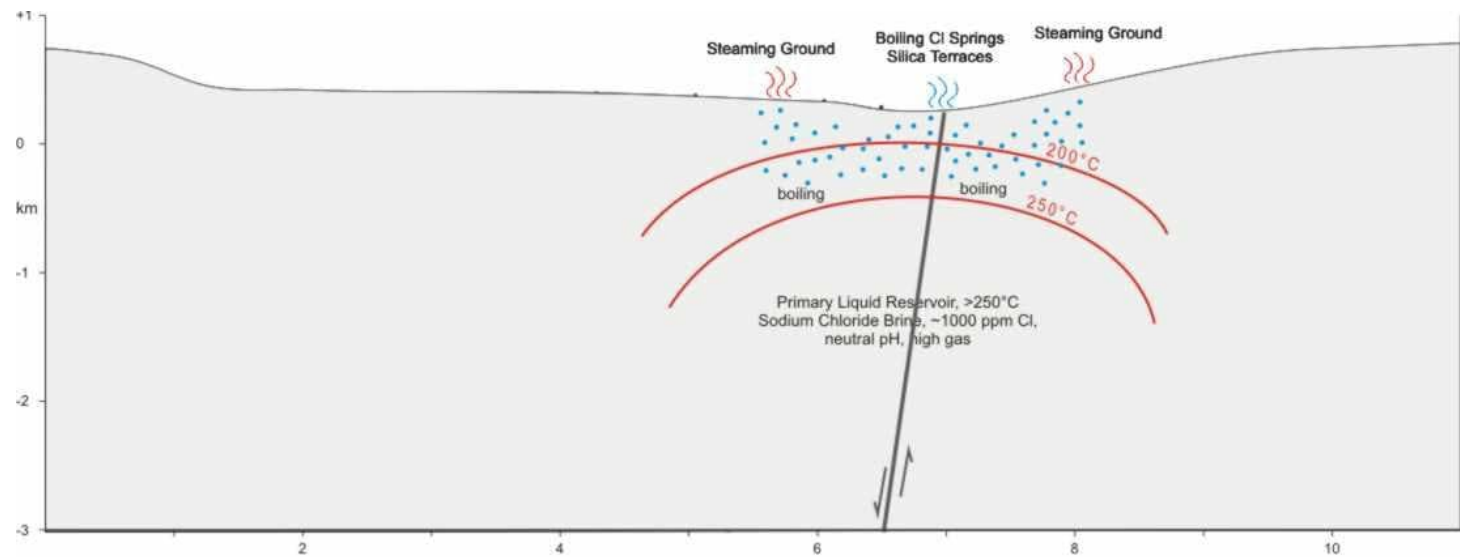
Chemical sampling – Geothermometry



Spring compositions plotted on the Na-K-Mg geoindicator diagram of Giggenbach (1988)

Conceptual Model. After Surface Reconnaissance (Geology and Chemistry)

- ❑ Major geological structures
- ❑ Indicated reservoir character
 - ❑ temperature
 - ❑ Chemistry
 - ❑ Reservoir extent



GEOPHYSICS:

➤ Magnetotelluric (MT):

- Survey design
- Calibration (instrument & sensor)
- MT data acquisition
- MT data processing
- 3D inversion & Interpretation

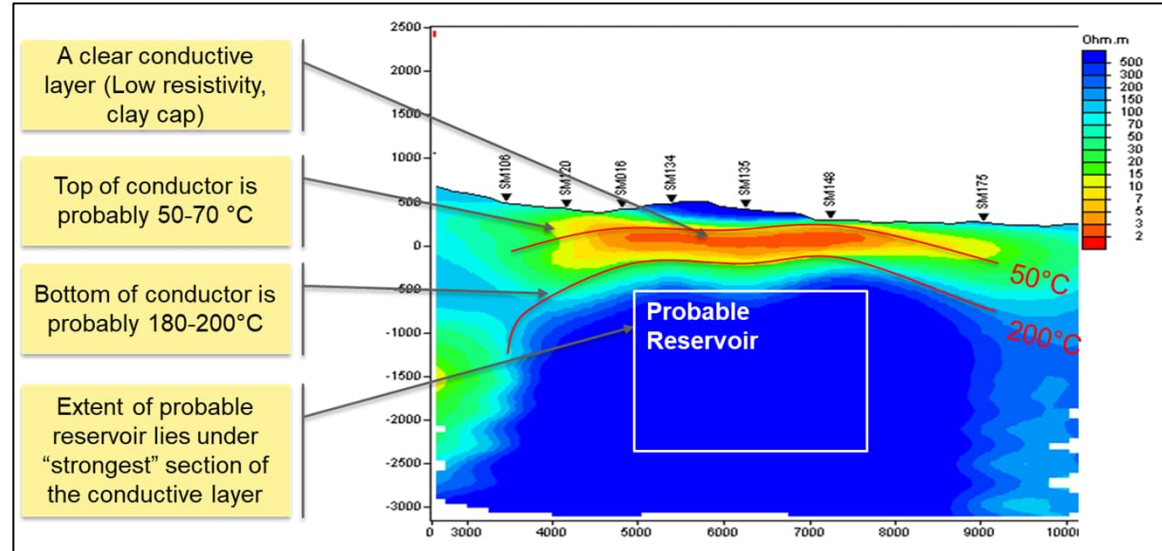
➤ Gravity:

- Survey design
- Calibration & Cycling
- Gravity data acquisition (Obs.)
- Processing
- CBA, RA & RS
- Modelling & Interpretation

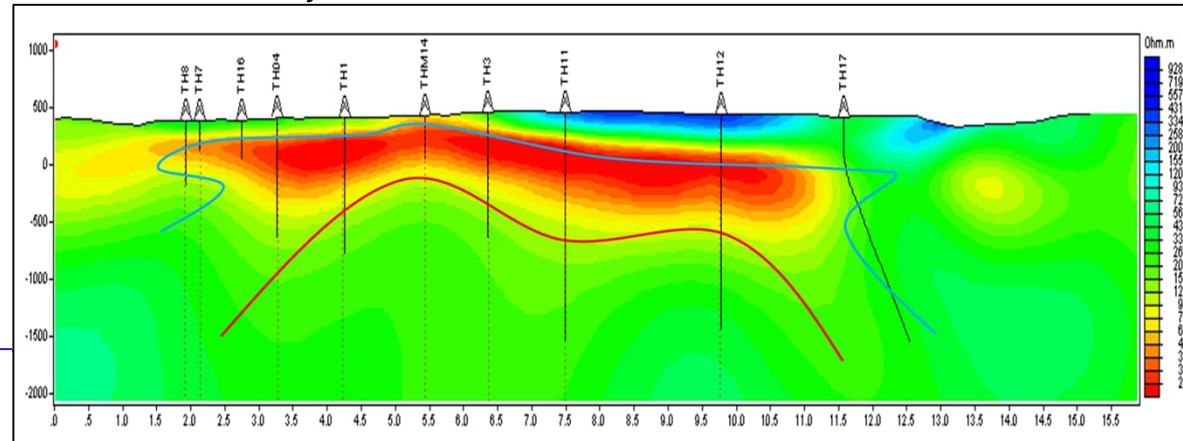
Outcomes:

- Top of Reservoir
- Reservoir Boundary
- Structure Indication

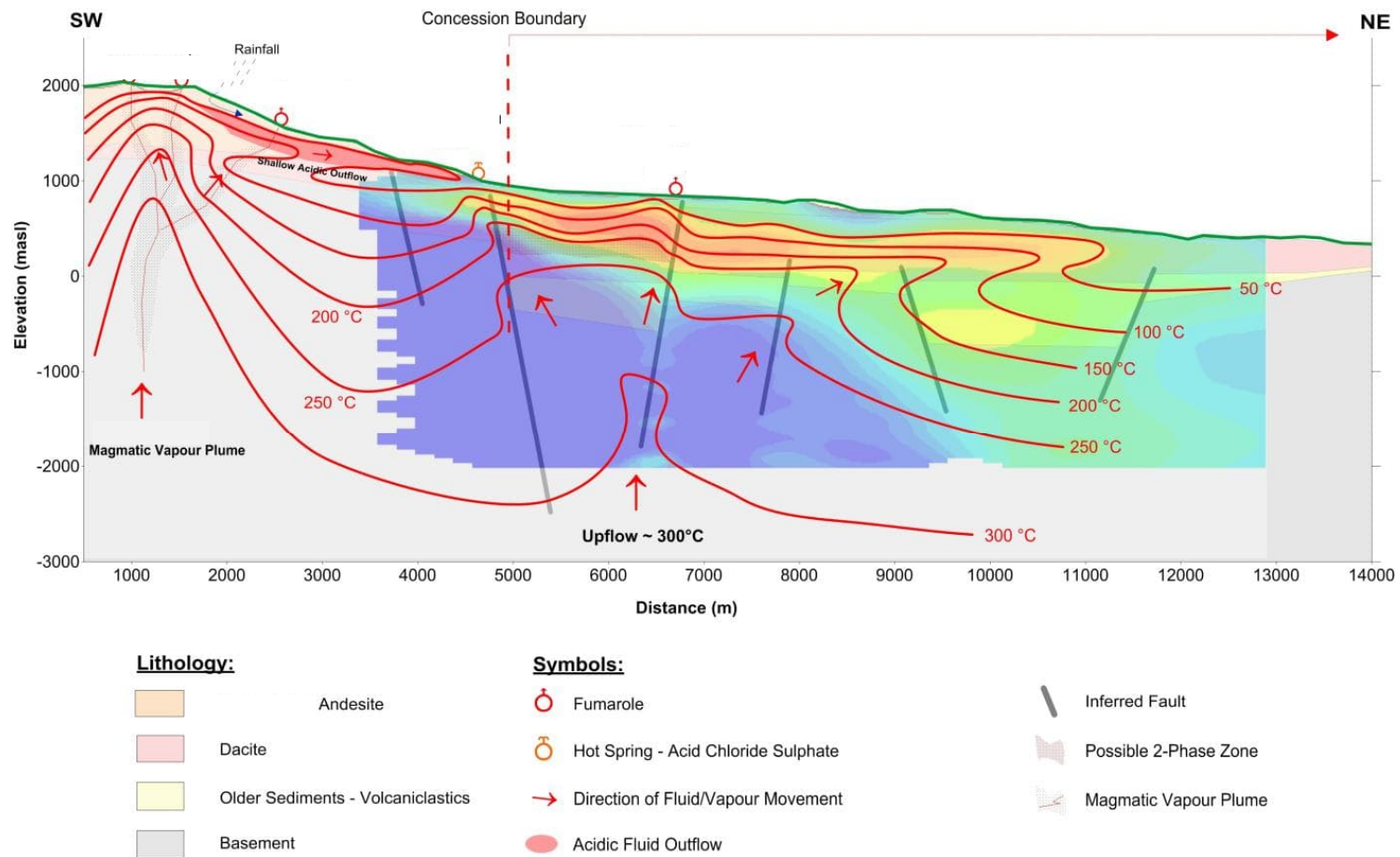
Geophysics help image the reservoir (look below the surface)



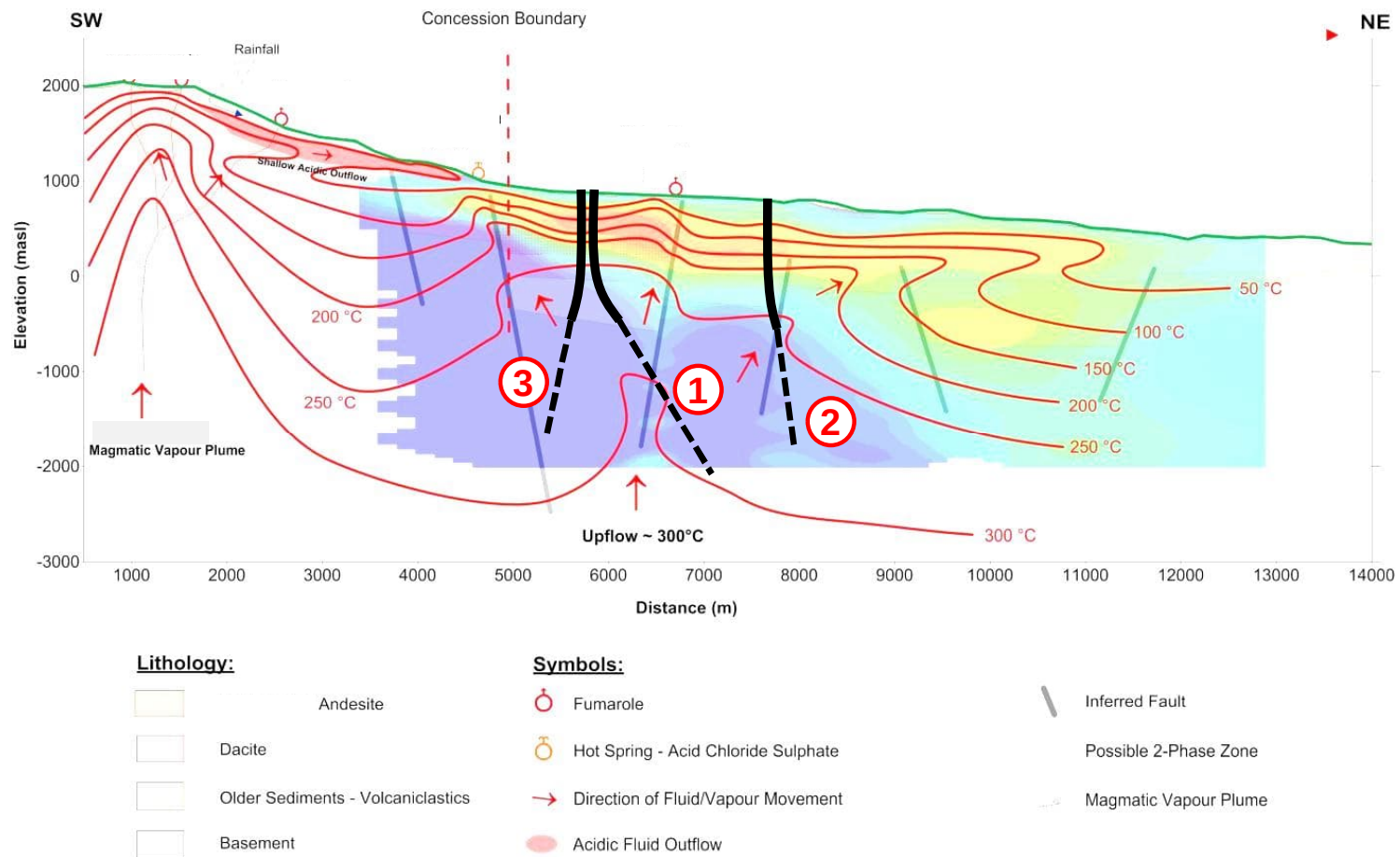
MT validated on many fields



Final Conceptual Model



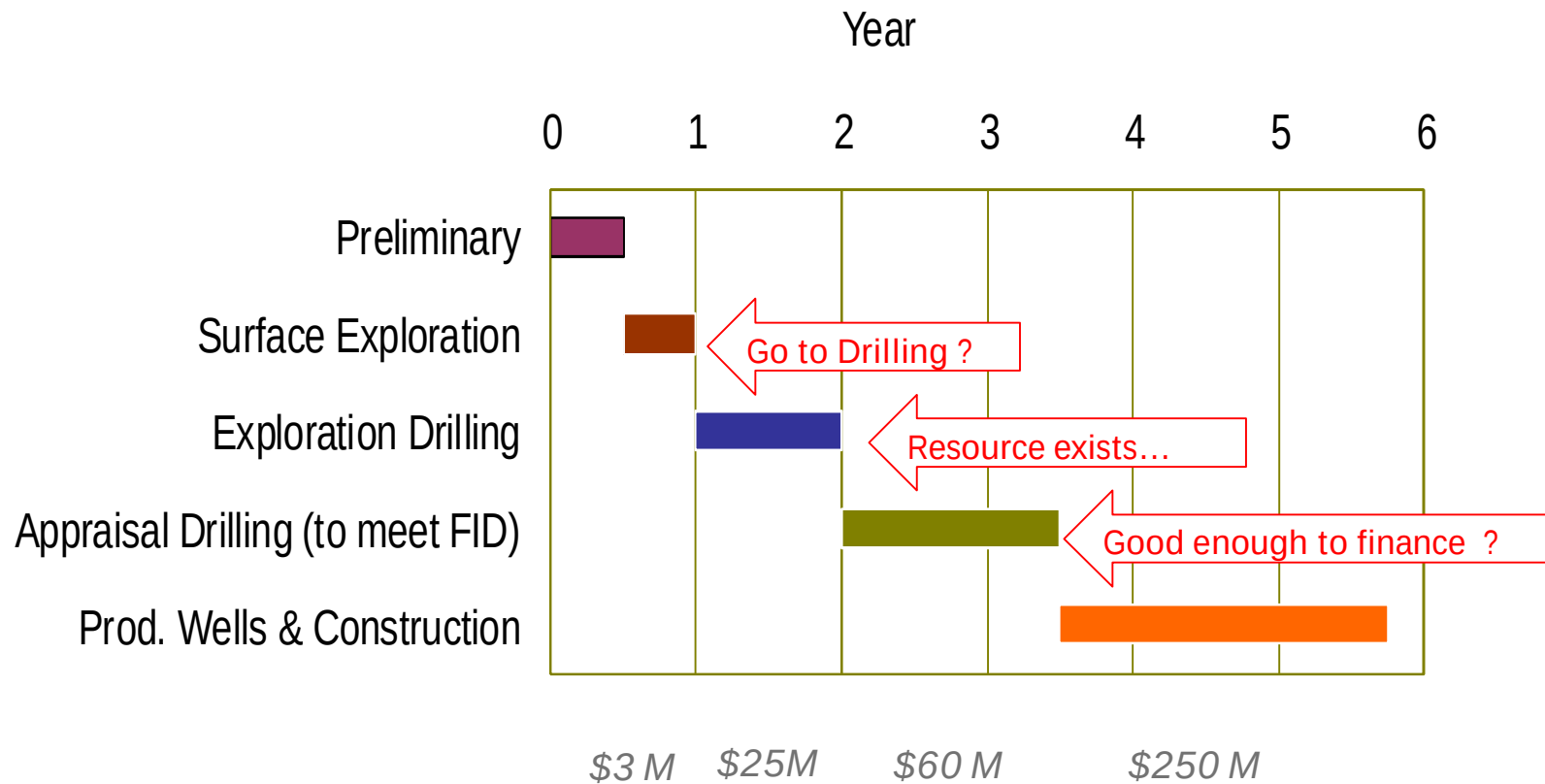
Exploration drilling program



Geothermal Project Development Steps

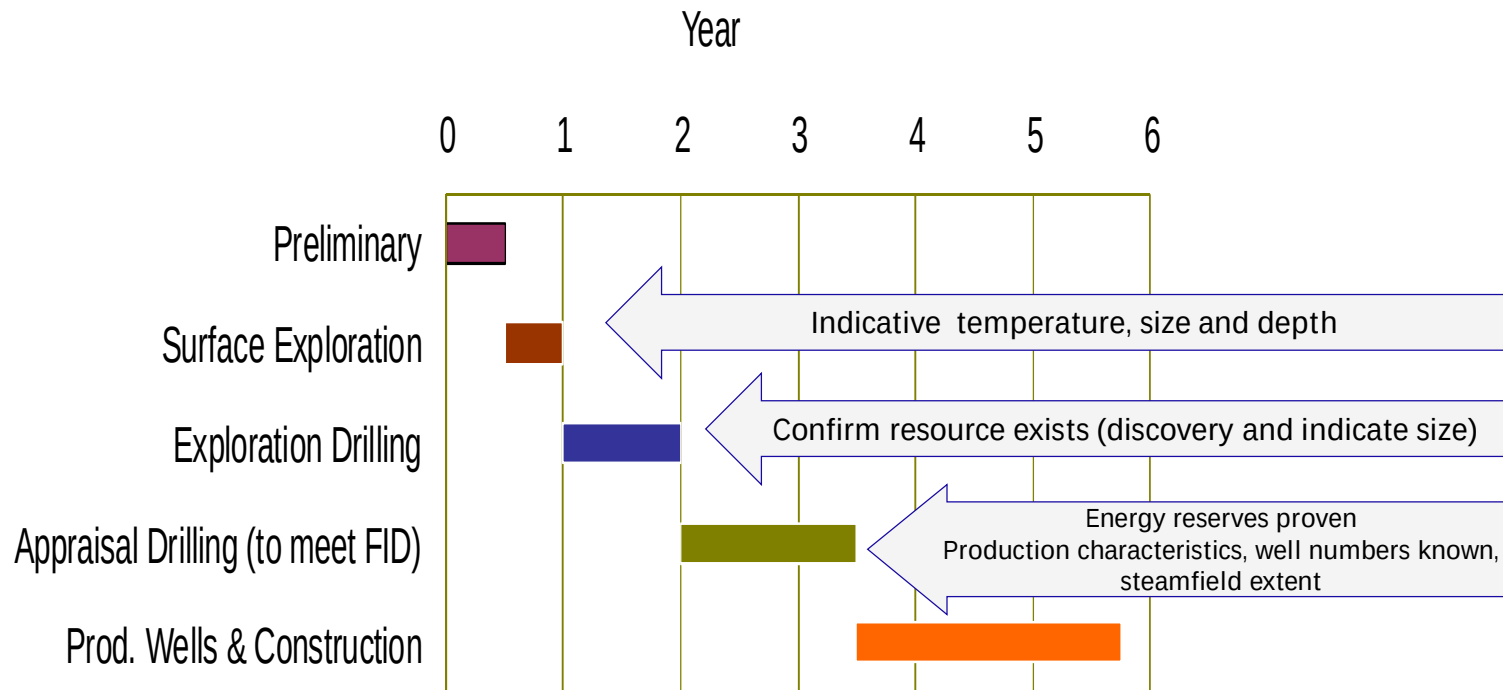
Exploration, Feasibility Study, FID, COD

Key decision points, and opportunity for delay...



Values indicative for a 50 MW project

Exploration and Drilling will progressively improve understanding of the fuel supply



Geothermal development phases in pictures

Development phases & timing for a greenfield geothermal project

	D1 ▼	D2 ▼	D3 ▼	FID ▼	COD ▼	EOL ▼
	Concession Acquisition	Scientific Fieldwork	Exploration Drilling	Confirmation Drilling, FEED & BFS	Construction & Development Drilling	Operation & Makeup Drilling. Expansion?
						
	Confirming market of electricity, price path, legal certainty on heat and water	Identifying that a geothermal resource is likely to exist & selecting discovery well targets	Proving a geothermal resource exists and proving/delineating sufficient reserves for development	Proving that development of the resource is technically and economically feasible	Constructing the geothermal facility on specification, on time and on budget	Operating and maintaining the geothermal facility and ensuring there is always enough steam

Note that these are distinct phases

D = Decision
FID = Financial Investment Decision
COD = Commercial Operation Date
EOL = End of Life

GO/NO-GO Key questions

- **Pre-Scientific Exploration Phase – Is it worth doing serious 3G/6G?**
 - Is there a market? Is there a price path?
 - **Output: Concept Study and budget for 3G/6G (D1)**
- **Scientific Exploration Phase - Is it worth doing exploration drilling?**
 - Conduct 3G
 - Are the 3G result promising?
 - Update Conceptual Study **to Pre-FS**
 - Prepare budget for Exploration drilling, include budget for infrastructure
 - **Output: Pre-FS and budget for Exploration drilling phase (D2)**
- **Exploration Drilling Phase - How to discover a Resource & Identify Sufficient Volume?**
 - Procurement for exploration drilling
 - Exploration drilling (discover & delineate) – Slimhole or full size?
 - How the exploration data and test result?
 - What is the size of project?
 - Update Pre-FS to **FS** based on exploration drilling result
 - Prepare budget for Confirmation Drilling & FEED
 - **Output: FS & budget for FEED & Confirmation Drilling**

Development phases & timing for a greenfield geothermal project

	D1	D2	D3	FID	COD	EOL
Concession Acquisition	Scientific Fieldwork	Exploration Drilling	Confirmation Drilling, FEED & BFS	Construction & Development Drilling	Operation & Makeup Drilling. Expansion?	
						
Confirming market of electricity, price path, legal certainty on heat and water	Identifying that a geothermal resource is likely to exist & selecting discovery well targets	Proving a geothermal resource exists and proving/delineating sufficient reserves for development	Proving that development of the resource is technically and economically feasible	Constructing the geothermal facility on specification, on time and on budget	Operating and maintaining the geothermal facility and ensuring there is always enough steam	
Concept study	Pre-FS	FS	Bankable FS FEED			

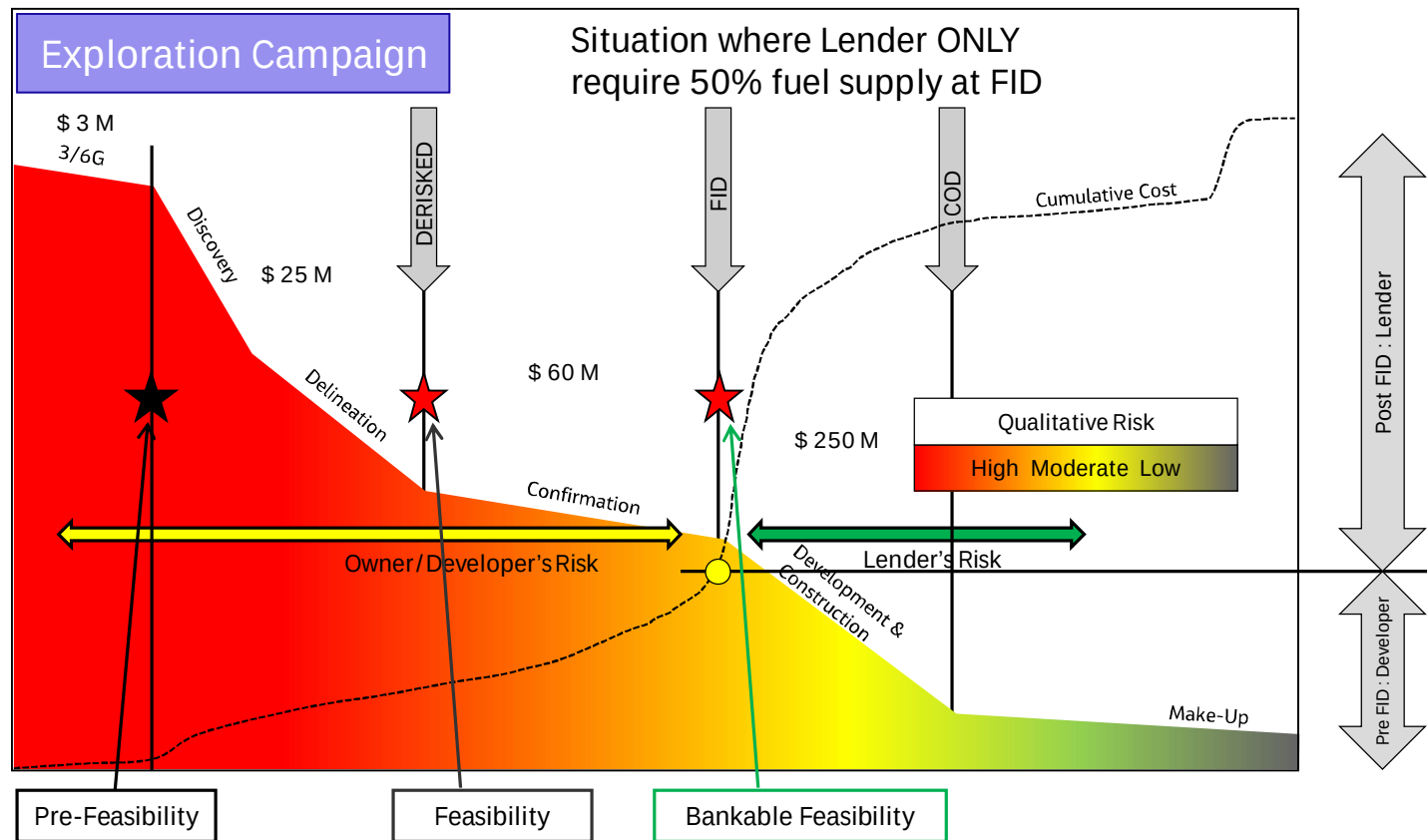
GO/NO-GO questions

- Confirmation Drilling & FEED Phase - What is needed to get funding for construction?
 - Permitting & Bankable Feasibility Study
 - Confirmation Drilling (what % under wellhead?) 50%? 100%
 - FEED & Procurement & Funding Agreements
 - Construction contract established
 - PPA established
 - Output : Bankable FS (BFS) FID/D4
- Construction Phase – Successfully build, test & commission the Facility
 - Development Drilling to get to (say) 120% of capacity at wellhead
 - Power plant, fluid collection and disposal system, transmission line, operational infrastructure
- Operations (& Maintenance) Phase – Maintain the Facility & keep it Steam Full
 - Avoid fuel supply constraints (geoscience, reservoir engineering and drilling)
 - Avoid surface facility constraints (availability and capacity, FCRS and PP)

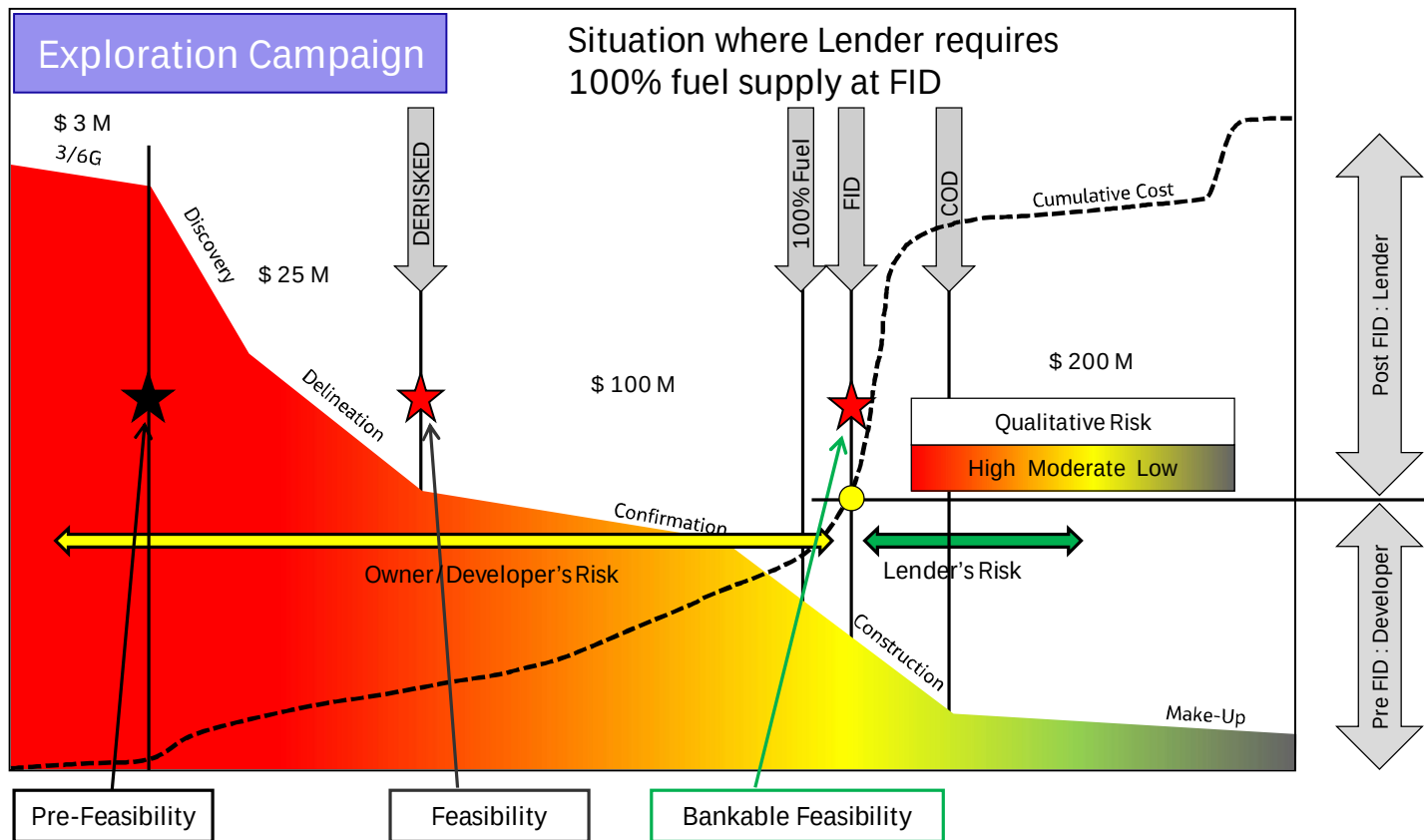
Development phases & timing for a greenfield geothermal project

	D1	D2	D3	FID	COD	EOL
	Concession Acquisition	Scientific Fieldwork	Exploration Drilling	Confirmation Drilling, FEED & BFS	Construction & Development Drilling	Operation & Makeup Drilling. Expansion?
						
	Confirming market of electricity, price path, legal certainty on heat and water	Identifying that a geothermal resource is likely to exist & selecting discovery well targets	Proving a geothermal resource exists and proving/delineating sufficient reserves for development	Proving that development of the resource is technically and economically feasible	Constructing the geothermal facility on specification, on time and on budget	Operating and maintaining the geothermal facility and ensuring there is always enough steam

Will lenders take fuel supply risk?
Here is what YES looks like



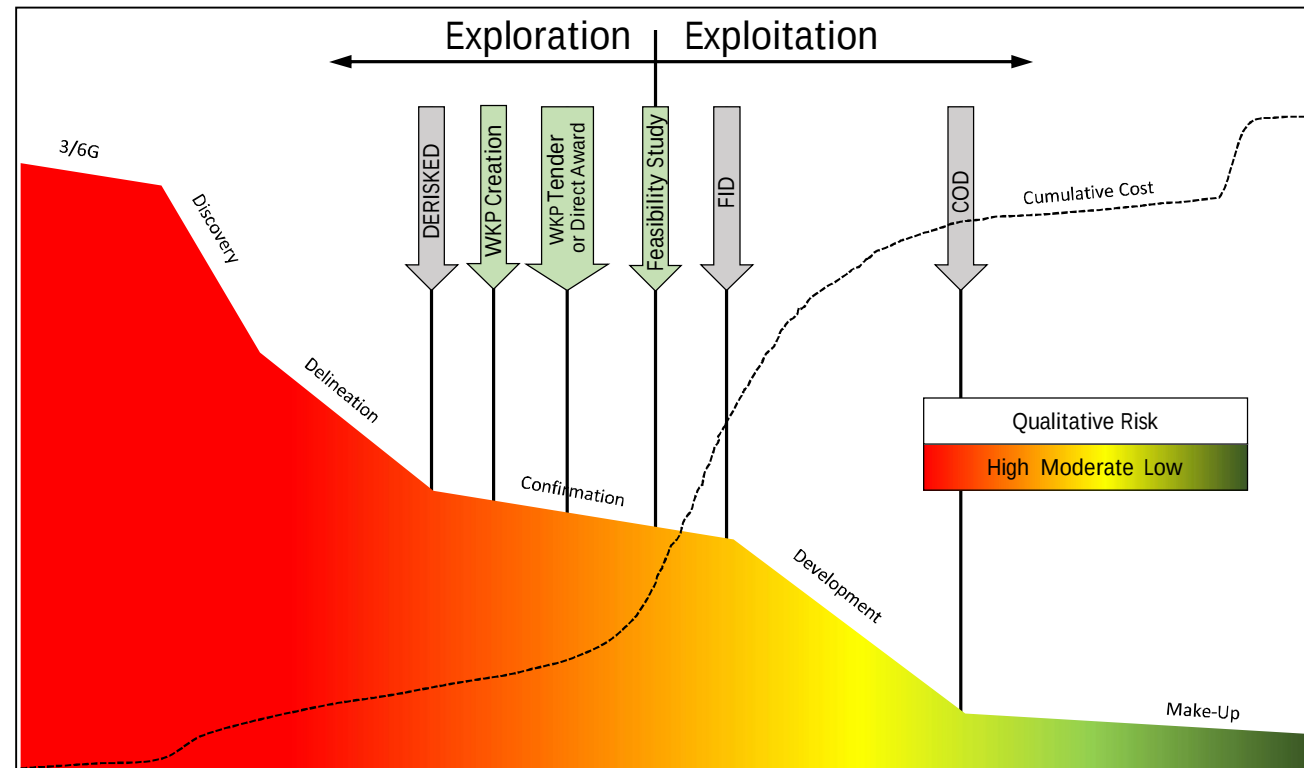
Will lenders take fuel supply risk?
Here is what NO looks like



Regulatory Certainty in Indonesia

Indonesia Geothermal Law
(UU 21/2014)

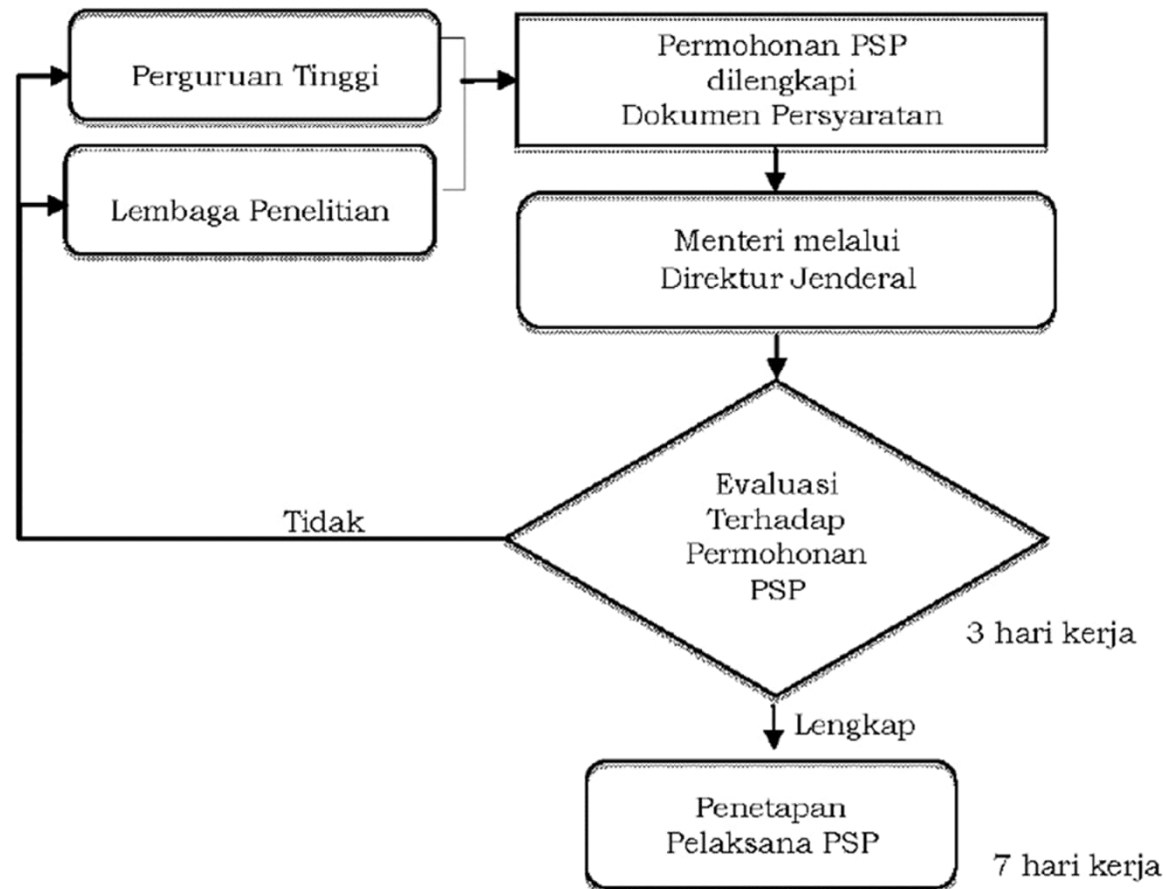
Milestones in the new law



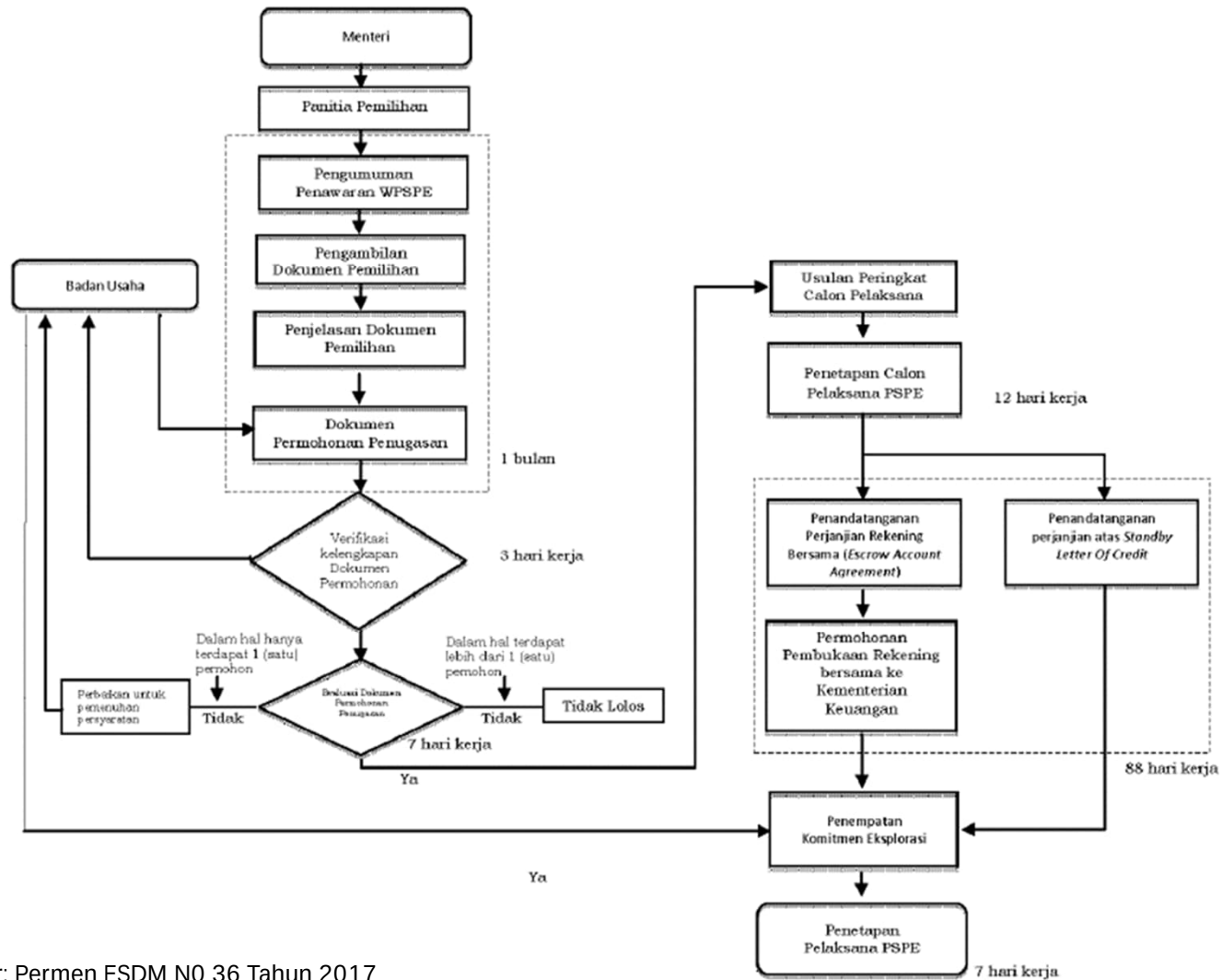
Regulatory Phase		Exploration			Exploitation						
Assignment to a Business Entity	Geothermal Area is in the hands of Government PENAWARAN WILAYAH KERJA				Geothermal Area is in the hands of the IPB Holder (non-SOE Business Entity)						
	Business Entity Carrying it Exploration on behalf of the Minister	PSP (Preliminary Survey by the Minister or on behalf of the Minister. This carried out in a defined PSP area (WPSP))	PSPE (Preliminary Survey and Exploration Drilling by the Minister or on behalf of the Minister. This carried out in a defined PSPE area (WPSPE))								
Assignment to a SOE	ALUR PENUGASAN WKP KEPADA BLU/BUMN				Geothermal Area is in the hands of the IPB Holder (BUMN/SOE Business Entity)						
Common Activities for WKP and post IPB award			WKP (Create and Proclaim a Geothermal Working Area)	Full Tender (if Govt did PSPE) or Limited Tender (to PSPE holder) or Direct Assign (to SOE) the WKP	An IPB is issued (Concession Licence)	IPB Holder completes exploration (per UU21/2014 and GR7/2017). FS is Submitted and Accepted by Minsiter	Confirmation drilling (to meet commercial requirements)	Production drilling (to meet commercial and technical requirements)	Operations (including reservoir management, makeup well drilling, facility O&M)	Expansion	
						Existing Development Area	New Development Area within the same WKP				
Regulatory Oversight	Regular reporting to the Minister (through EBTKE)				Regular reporting to the Minister (through EBTKE)						
Activity	3G	3G (infill or complete), and drilling (slim or std.), at least one deep well, Pre-FS	WKP creation	Tender or assignment process	Additional drilling (if required), (slim or std.), Prepare FS,	Confirmation drilling (if Req'd.) for financing, Bankable FS, NOID, PPA	Production drilling, construction	O&M, makeup drilling	Additional 3G, drill step-out wells, FS for additional units(s)	Additional 3G, drill exploration wells, FS for additional units(s)	

Common Activities for WKP and post IPB award	PSP	PSPE	WKP (Create and Proclaim a Geothermal Working Area)	Full Tender (if Govt did PSPE) or Limited Tender (to PSPE holder) or Direct Assign (to SOE) the WKP	An IPB is issued (Concession Licence)	IPB Holder completes exploration (per UU21/2014 and GR7/2017). FS is Submitted and Accepted by Minsiter	Confirmation drilling (to meet commercial requirements)	Production drilling (to meet commercial and technical requirements)	Operations (including reservoir management, makeup well drilling, facility O&M)	Expansion	
									Existing Development Area	New Development Area within the same WKP	
Regulatory Oversight	Regular reporting to the Minister (through EBTKE)					Regular reporting to the Minister (through EBTKE)					
Activity	3G	3G (infill or complete), and drilling (slim or std.), at least one deep well, Pre-FS	WKP creation	Tender or assignment process		Additional drilling (if required), (slim or std.) Prepare FS,	Confirmation drilling (if Reqd.) for financing, Bankable FS, NOID, PPA	Production drilling, construction	O&M, makeup drilling	Additional 3G, drill step-out wells, FS for additional units(s)	Additional 3G, drill exploration wells, FS for additional units(s)
Range of resource outcomes at end of phase		One deep well through to all steam required for COD				One deep well through to all steam for COD	Resource proven, and production proven for financing				
Offtaker Considerations (for sale of electricity to PLN)		PLN written confirmation of Initial Development capacity	A positive mention in the RUPTL			NORC issued and accepted by PLN. Form of PPA confirmed by both Parties.	PPA executed	Both Parties meet PPA commitments		PPA negotiated	PPA negotiated
Milestone to be achieved at end of phase, including typical project milestones		Resource "discovered" (per UNFC)	WKP gazetted	IPB issued		Resource proven, FS accepted	FID (Financial Investment Decision)	COD (Commercial Operation Date)	Hand back	FS accepted	FS accepted
Possible failure		3G (PSP) and/or several wells show: - no / poor resource - not commercial	RUPTL does not support development	Bid fails		Resource is not sufficient Non commercial	The project is not: - commercially / technically feasible	The project is : - late, costs more, below capacity	The project has: - has low capacity		
Failure outcome		Walk away after 3G Walk away after some drilling		No bid		Walk away after some drilling	On hold/walk away	Diminished returns	Diminished returns		
Possible duration	1+1 years	3+2 years	Not defined by regulation, but hoped to be less than 1 year			5+2 years	0-3 years	1-2 years	29-30 years		
							30 years				
Typical Timing (seq. year), 50 MW	N/A (assume no PSP)	3	4			5-6	6-8	7-10			

BAGAN ALIR PERMOHONAN PSP



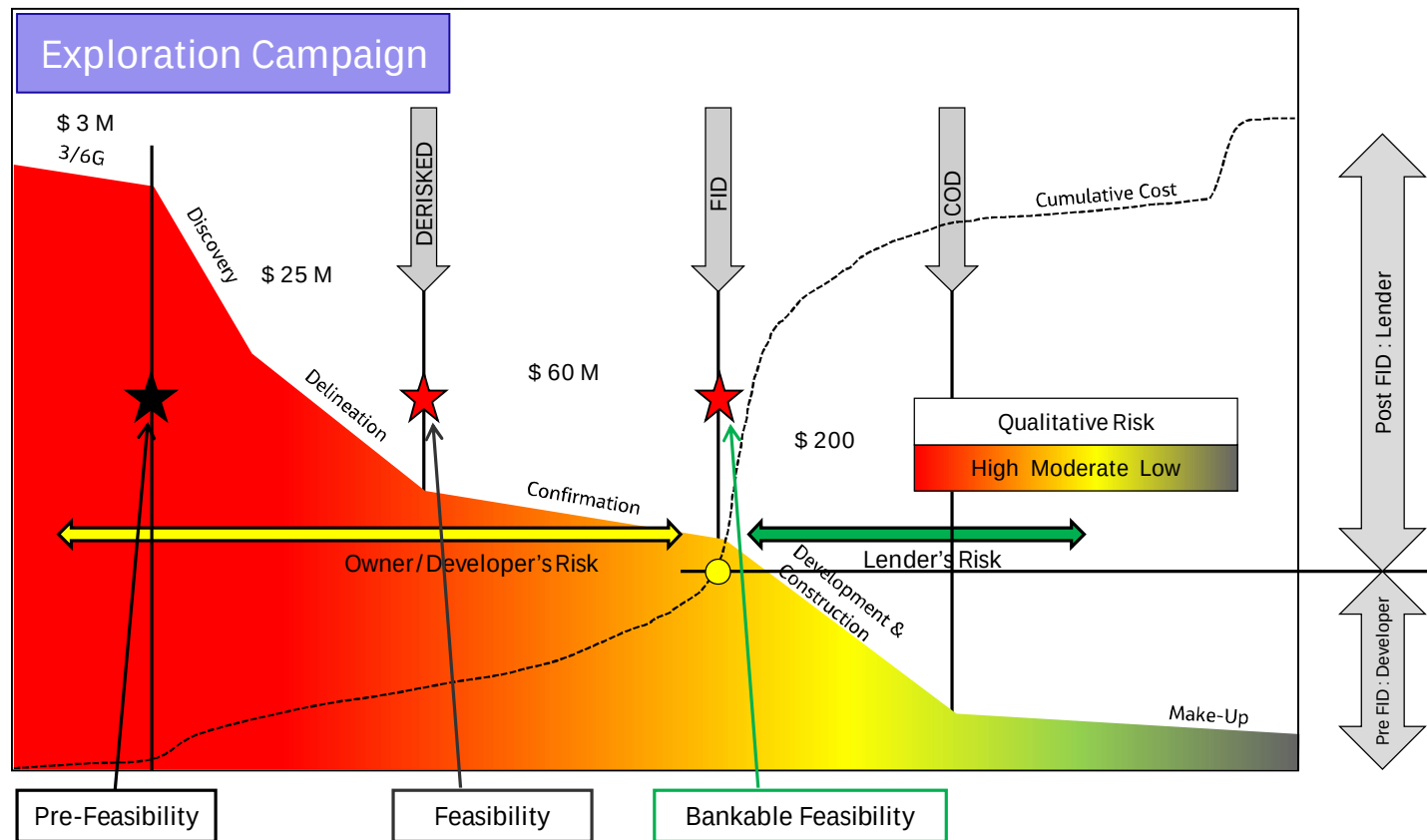
BAGAN ALIR PERMOHONAN PSPE



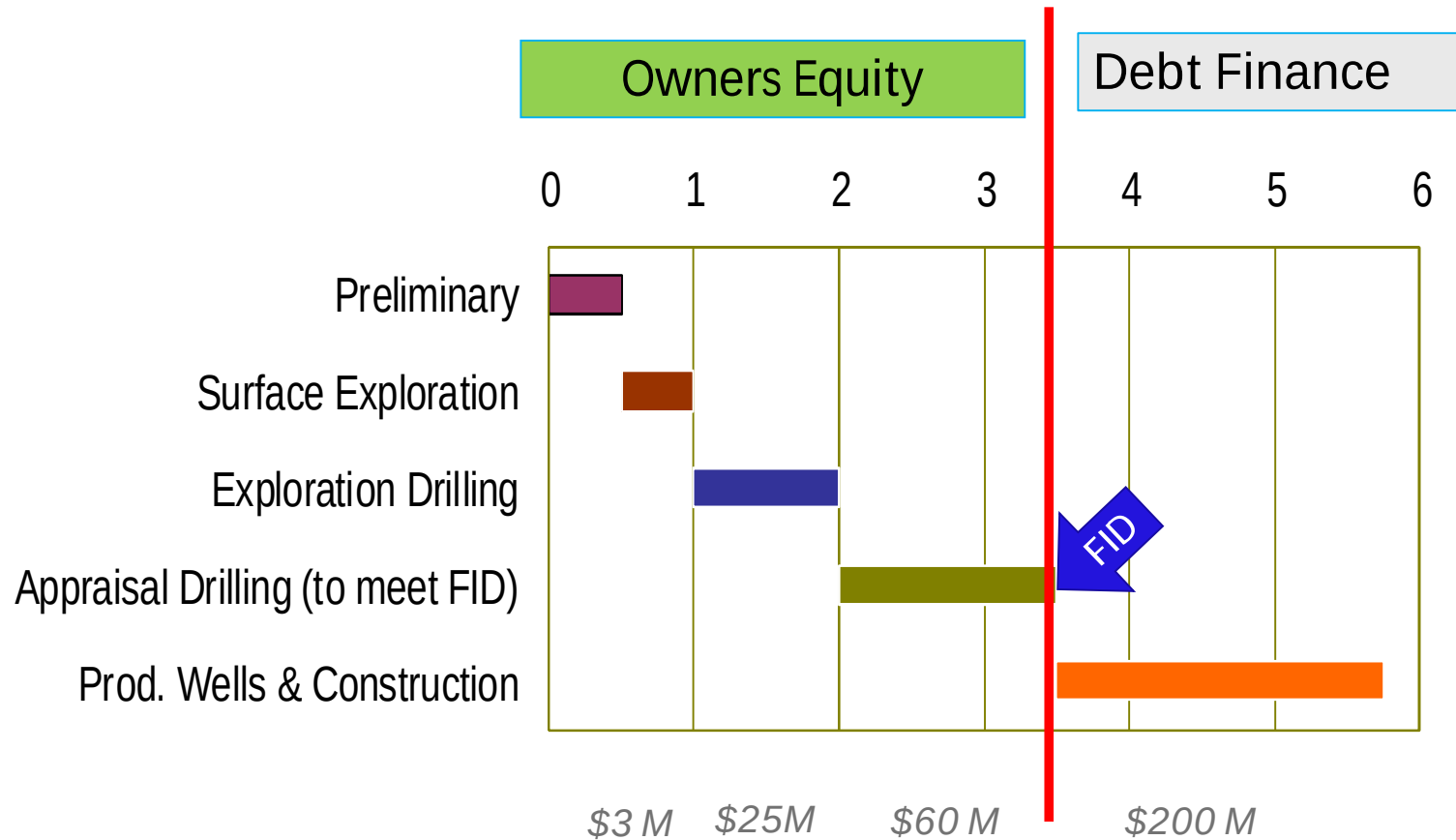
Indicative Cost

End to End CAPEX

Meet the Curve Again....



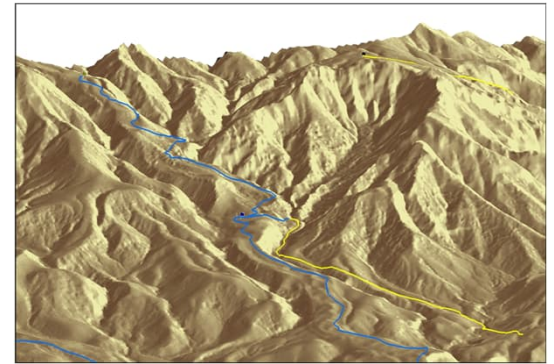
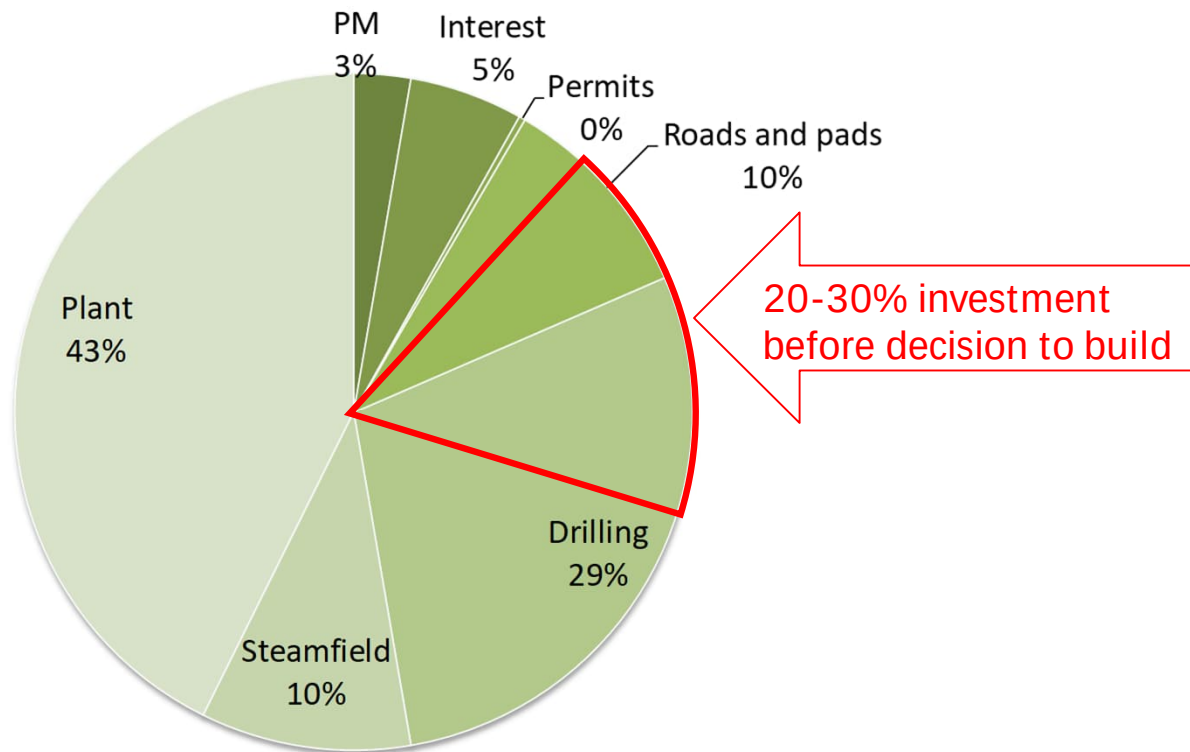
Indicative Cost for Each Phase



Indicative Cost – For a 55 MW Plant During Exploration

- Average Cost of drilling per well
 - USD 5-10 million per well (standard well)
 - USD 2-3 million per well (slim hole well for exploration)
 - Exploration :
 - Permen No 36 – 2017 requires at least 1 exploration well
 - Technically 2 exploration wells is normally used to get good understanding on how reservoir extend
 - Can be combination of 1 Slimhole and 1 std hole
 - During Exploration Drilling phase:
 - 3G/6G cost : US\$ 1-2 million
 - Civil infrastructure : US\$5-10 million*
 - Road access, bridges, laydown area, 3 wellpads
 - Drilling water supply (pump, temporary piping) : US\$2-3 million
 - Drill 2 Slimhole well : USD 4-6 million
 - If 1 Slimhole + 1 Std hole : USD 13 million
 - Others : Permit, operation cost, resource assessment review, contract
 - Total CAPEX around US\$20-30 million at this phase
-

High cost Up-Front



Indicative Cost – For a 55 MW Plant During Confirmation Drilling

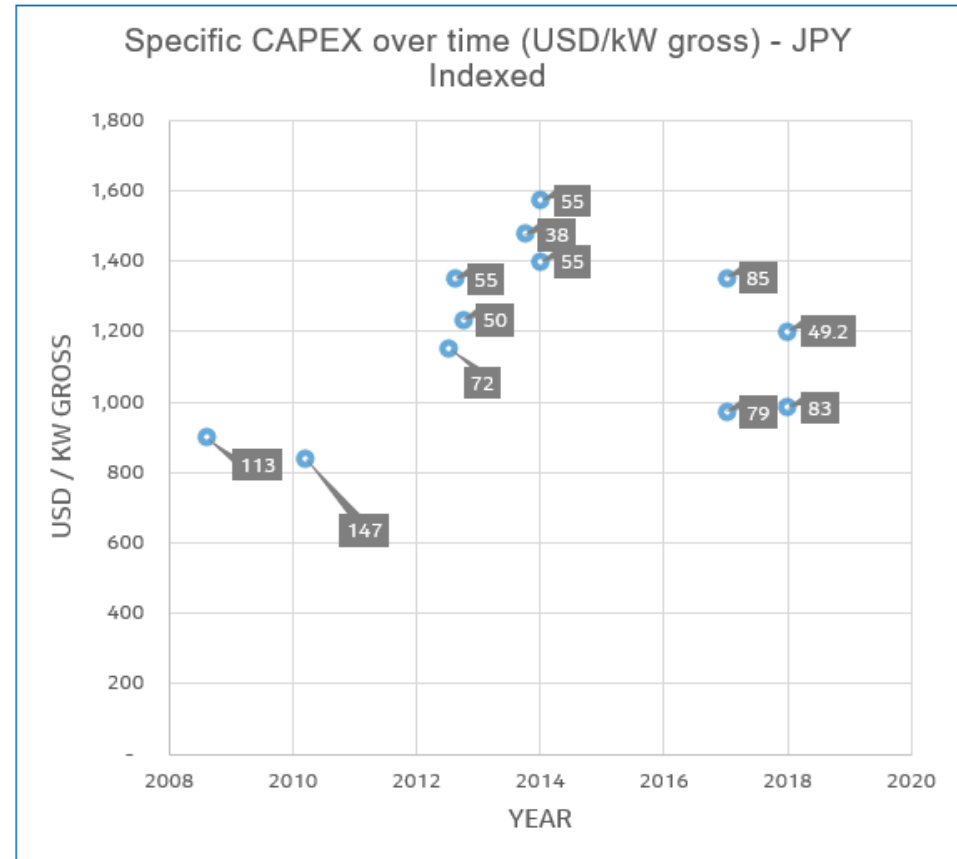
- During Confirmation Drilling phase:
 - More wells to be drilled. Each well is assumed 10MW per well.
 - Drilling success rate is 75% (with success exploration phase)
 - Assumed lenders require 50% proven well.
 - About 3-4 production wells will be drilled → USD 30-40 million
 - Need injection well as well → USD 10 million
 - Civil infrastructure
 - Well Test
 - Engineering : Drilling evaluation, update modelling
 - ESIA for Power Plant
 - Project management : Drilling & Civil Infrastructure
- Total CAPEX around US\$50-60 million at this phase

Indicative Cost – For a 55 MW Plant During Development Phase

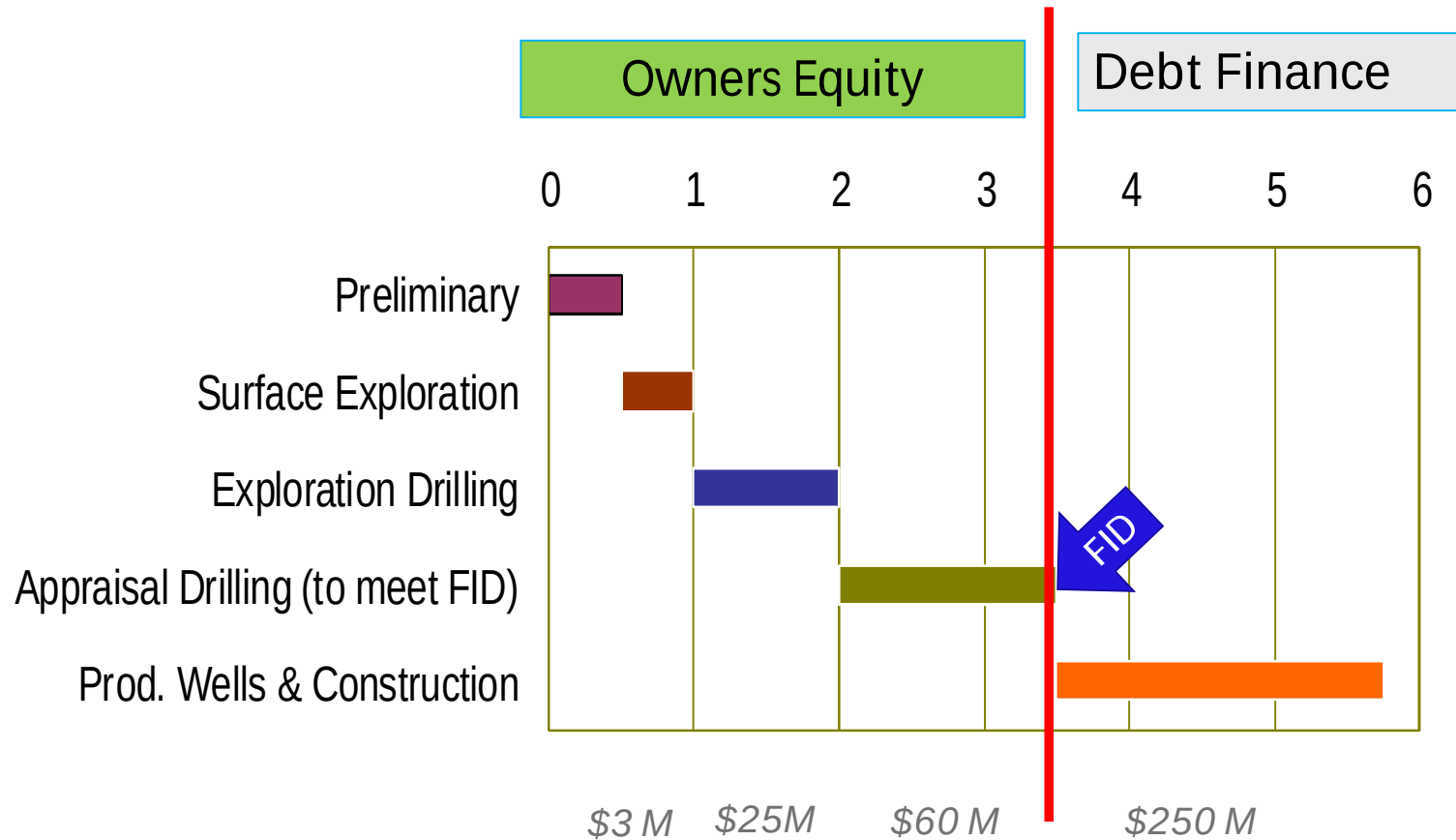
- During Production Drilling Phase:
 - More wells to be drilled. Each well is assumed 10MW per well. To achieve 110% capacity of COD
 - Drilling success rate is 80%
 - About 4-5 production wells will be drilled (std hole) → USD 40-50 million
 - Need injection well as well (1-2ea) → USD 10-20 million
 - Total CAPEX for Drilling at this phase around US\$60-70 million
- Steamfield (FCRS – pipeline, separators, vent station) : US\$20-30 million
- Engineering : Drilling evaluation, update modelling
- ESIA for Power Plant
- Project management : Drilling & Civil Infrastructure
- Total CAPEX around US\$80-110 million at this phase (excluding power plant)

Power Plant Cost

- At 55 MW, power plant cost is ~ USD 1500 / kW
- For 1x55MW plant, approximately power plant cost is US\$70-80 million
- 1x55 MW switchyard : US\$3-4 million
- Transmission line: US\$300,000 per km
- Approximate cost for Power Plant (exclude transmission) : US\$85 million



Indicative Cost for Each Phase



Values indicative for a 55 MW project

Unseen Cost

Examples of Unseen Cost :

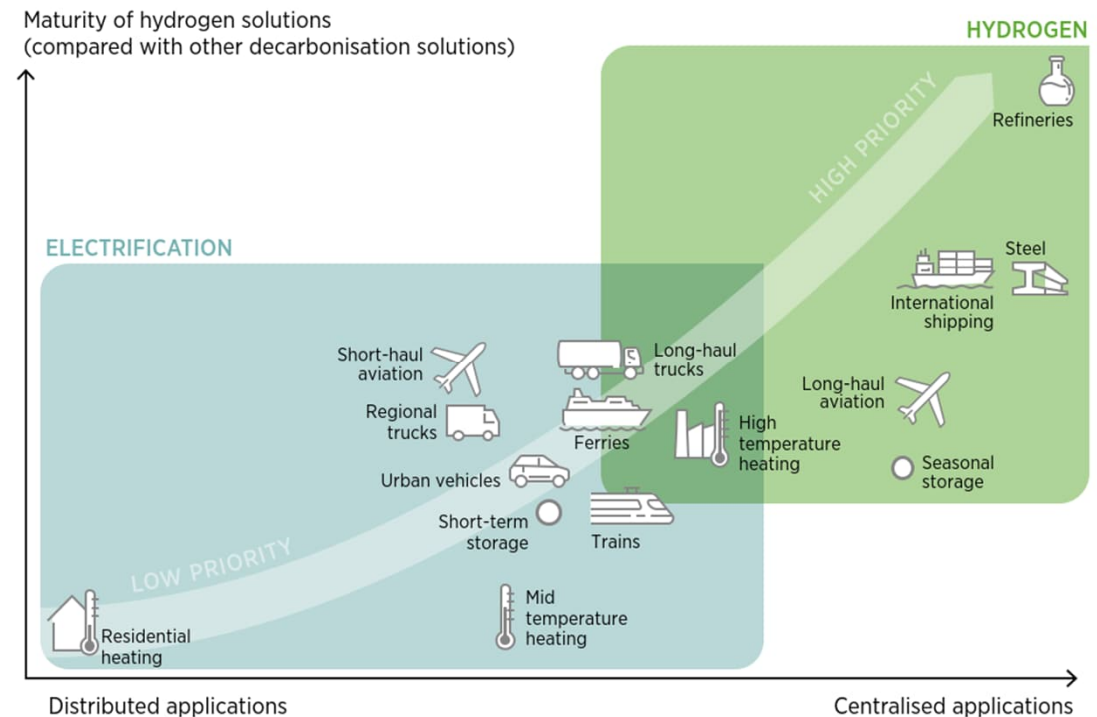
- When things go wrong
 - Exploration drilling find no geothermal fluid or drill in wrong place (this happens!)
 - Drilling problem (formation, rig issue) lead to longer drilling time
 - Project is smaller than we thought. Economic scale is reduced
- Community
 - May need to change plan (road, different pads)
 - Schedule delay
- Special purpose vehicle (SPV)
 - Indonesia law require one company to own WKP. In case where JV formed, one company need to be established
 - Hence overhead cost required
 - Schedule delay
- Permits
 - Need permitting plan
 - Environment, safety : Multiple stages sign-off (incl local government)

Beyond Geothermal Electricity

Green Hydrogen, Direct Use, Mineral Extraction

Why Hydrogen ? – A Role in Decarbonisation

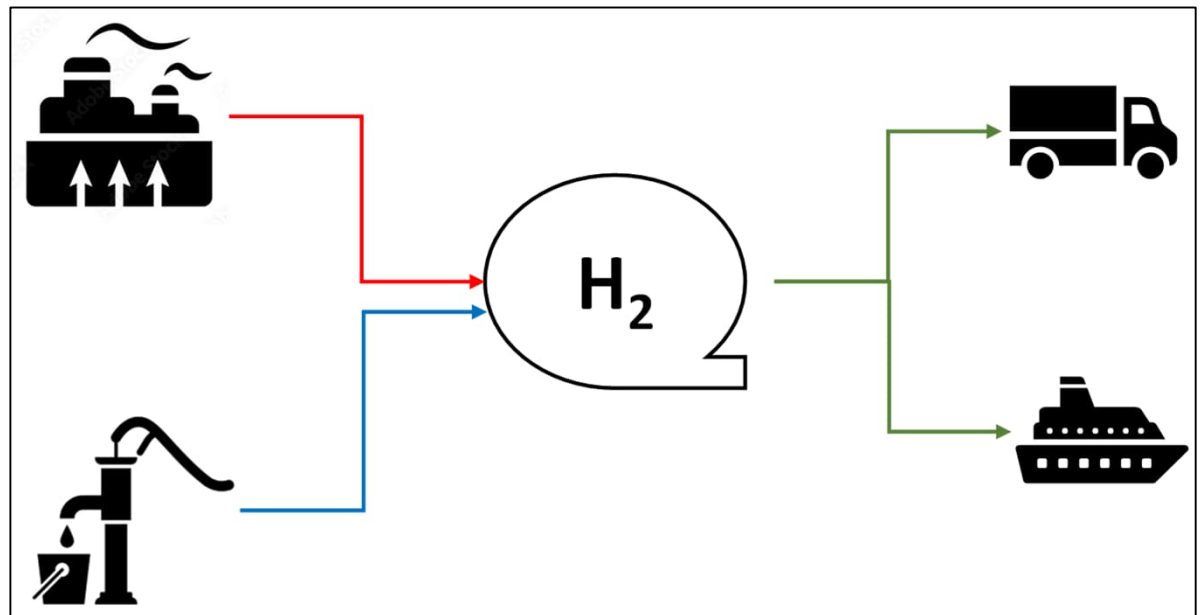
- Green Hydrogen = Hydrogen produced from renewable energy
- Green hydrogen plays important role in energy transition
- H2 mostly used in:
 - Refineries
 - Ammonia / Urea
 - Methanol
 - Biodiesel
- H2 can be used in:
 - Steel industry
 - Fuel Cell Electric Vehicle



Source: IRENA 2022

Potential Business from Geothermal Other than Electricity – Green Hydrogen

- Electricity generated by geothermal can be used for hydrogen generation (electrolysis)
- Compare to other green energy source, geothermal offer higher capacity factor (95%)
- Generally 1 MW ~ 141 ton/year of hydrogen produced
- Generally cost to produce green hydrogen from geothermal is USD 5-6 USD /kg
- Price of green hydrogen in market now : USD 6-7 /kg
- With this price, if 55 MW geothermal plant produce green hydrogen, it can generate revenue ~20% more than selling electricity only (assuming 8 cent/kWh tariff)



Potential Business from Geothermal Other than Electricity – Mineral Extraction

- Geo40 – Silica Extraction



Fasilitas Geothermal Resource Risk Mitigation (GREM)

GREM merupakan fasilitas pembiayaan eksplorasi panas bumi yang dapat diakses oleh sektor swasta dan publik di Indonesia. Kegiatan ini bertujuan untuk mengurangi risiko tahap awal pengembangan proyek melalui metode *de-risking* atau pembagian risiko. Fasilitas pendanaan ini disetujui oleh Green Climate Fund (GCF) pada tahun 2018 dan tercatat sebagai proposal pendanaan pertama yang disetujui di Indonesia. Total fasilitas ini sebesar USD 651,25 juta dengan sumber pendanaan dari GCF, International Bank for Reconstruction and Development (IBRD) World Bank, Clean Technology Fund (CTF), dan Kementerian Keuangan melalui PT Sarana Multi Infrastruktur (PT SMI). Durasi awal fasilitas ini selama 10 tahun dan akan selesai pada tahun 2030. Deskripsi lengkap mengenai fasilitas ini tersedia dalam [proposal yang sudah disetujui oleh GCF](#).

Fasilitas ini terdiri dari jendela pendanaan swasta dan publik, dengan penerima manfaat, skema, dan sistem pembiayaannya masing-masing. Per Juli 2021, World Bank sebagai lembaga terakreditasi dan PT SMI sebagai entitas pelaksana telah berkolaborasi dalam operasionalisasi kedua jendela tersebut.

Jendela Publik	Jendela Swasta
Di dalam jendela pendanaan publik, fasilitas GREM mendukung pengembang publik (BUMN/ anak perusahaan BUMN) melalui pemberian pinjaman multilateral (IBRD, GCF, dan CTF) dan penyediaan fasilitas <i>de-risking</i> dari Geothermal Sector Infrastructure Financing Fund (PISP Fund) dengan maksimum batas agregat sebesar USD 30 juta.	Melalui jendela pendanaan swasta, fasilitas GREM mendukung pengembang swasta melalui kombinasi pinjaman konvensional dan sistem langganan dengan instrumen keuangan (financial instrument/FI) yang diterbitkan oleh pengembang (berbasis 50:50). Pinjaman konvensional berasal dari pinjaman IBRD dan sistem langganan untuk FI berasal dari Reimbursable Grant GCF dan/ atau Contingent Recovery Grant CTF.

Implementasi Perlindungan Lingkungan dan Sosial (Environmental and Social Safeguards/ESS) & Gender

Implementasi kebijakan sesuai dengan laporan ESS. Kebijakan ini akan dinilai kembali setelah penyaringan setiap sub-proyek dilakukan. Fasilitas GREM telah mengembangkan [Environmental and Social Management Framework \(ESMF\)](#), yang dapat diakses melalui situs web [PT SMI](#) dan World Bank sejak Mei 2019.

Rencana Tindak Lanjut

- Tindak lanjut diskusi dengan OJK mengenai penyalarsan instrumen pendanaan GREM melalui jendela sektor swasta dengan POJK 30/2019.
- Memantau perkembangan revisi PMK 62/2017 yang dapat mempengaruhi skema pendanaan GREM dalam jendela sektor publik.
- Peningkatan pemahaman calon pengembang terkait fasilitas GREM dan persyaratannya. Sebuah [manual bagi pengembang](#) yang memberikan informasi terkait manfaat, persyaratan, dan proses pengajuan telah tersedia melalui situs web PT SMI.

Linimasa Fasilitas GREM



Lembar fakta ini dikembangkan berdasarkan laporan tahunan yang dikumpulkan oleh Entitas Terakreditasi kepada GCF. Informasi yang dibagikan ini telah diulas oleh Entitas Terakreditasi dan Entitas Pelaksana proyek GREM.

Kontak:
ndagcf-indonesia@kemenkeu.go.id

Terimakasih

Jacobs

Challenging today.
Reinventing tomorrow.

